

[5871]-737

B.E. (Information Technology)
DISTRIBUTED COMPUTING SYSTEM
(2015 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q. 10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain in brief key techniques used for indirect communication in Distributed system. [5]

b) What are various forms of transparency in distributed system? Illustrate network transparency with an example. [5]

OR

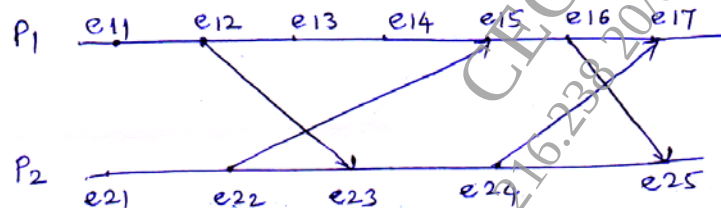
Q2) a) What is group communication? Discuss IP multicast w.r.t. following : [5]

- i) Multicast Routers
- ii) Multicast address allocation

b) Describe how connectionless communication between a client and a server proceeds when using sockets. [5]

Q3) a) Explain the central server algorithm and evaluate the performance of this algorithm for mutual exclusion. [6]

b) Compute Lamport's clock's logical event ordering for the following scenario: [4]



OR

- Q4)** a) Distinguish between caching and replication. [4]
b) Using finite state diagrams for the coordinator and the participant, describe the two-phase commit protocol. [6]

- Q5)** a) List the different distributed file system requirements. Explain the abstract file service architectural model with neat diagram. [8]
b) Explain the principal design feature of Bittorrent file sharing application, and explain in detail how Bittorrent protocol operates. [8]

OR

- Q6)** a) What is meant by Directory Services? What are Directory service operations. [8]
b) Discuss Domain Name System with it's components like DNS queries, DNS name server. [8]

- Q7)** a) Describe diagrammatically the general organization of the Apache Web server. [8]
b) What are web server clusters? Explain the content - aware cluster of web Servers with a neat diagram. [8]

OR

- Q8)** a) Give the disadvantage of using Hierarchical caches for a web proxy. How can it be overcome through cooperative caching? [8]
b) Why web services are used? Explain SOAP and REST based web services. [8]

- Q9)** a) List out the various security threats in a distributed system. Elaborate on the various policies and security mechanisms in distributed systems. [9]
b) Describe KERBEROS authentication system architecture with neat diagram. Explain why KERBEROS is not complete security solution. [9]

OR

- Q10)** a) Explain the concept of Grid. Explain in brief, the requirements of Grid applications. Distinguish between Grid and Coloud computing. [9]
b) What is meant by Public key cryptography? Explain Digital signatures with public keys. [9]

